

Database tables 5m

How do databases organise the data?

Databases organise their data in tables. A relational database can be regarded as a collection of tables consisting of rows and columns filled with data (much like a workbook consisting of multiple spreadsheets). There is a relation between the table's metadata, which is what specifies the valid type and range of data in each column, and the data itself. Each row in the table, also known as a tuple, represents a data record, and each record in a table has the same structure, defined by the column headings. Each item of data (where a row and column meet) is known as a field.

The image below demonstrates what metadata, columns, tuples/rows and fields are in a relational database.



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Metadata, columns, tuples/rows and fields are in a relational database

Column 1	Column 2	Column 3	Column 4	Column 5

Metadata Column Row/tuple Field

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Both the rows and columns in a relational table are sets, which means that they are unique and occur in no particular order, making it impossible to retrieve the 'first row' or the 'last row' in a relational table. It is only possible to retrieve rows by specifying their content; for example, 'row with account number 1023041' or 'all rows with the date of birth field before January 1 1998'. That, at least, is the theory.

In practice, actual implementations of relational databases will have a physical order of columns and rows, but you should not make assumptions about this order or try to use it in your interactions with the database. The combination of the database and the tools associated with managing its data is known as a relational database management system (RDBMS).

Dig deeper

Refer to *Database Design Basics* (Microsoft, n.d.) as a helpful resource for learning more. It's a straightforward tutorial focused on database design, particularly emphasising what data should go into columns. Note that it explains the design process without using an Entity Relationship Diagram (ERD).

Next we'll unpack database schema.



Thoughts on this page?

